Chapter 7, Problem 5RQ

Problem

In the circuit of Fig. 7.79, the capacitor voltage just before $t = 0$ is:

(a) 10 V

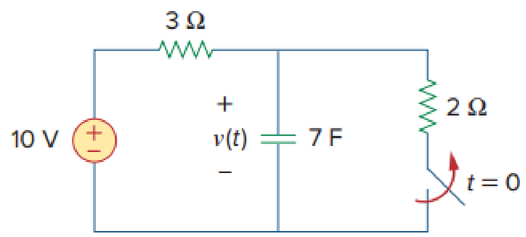
(b) 7 V

(c) 6 V

(d) 4 V

(e) 0 V

Fig. 7.79:

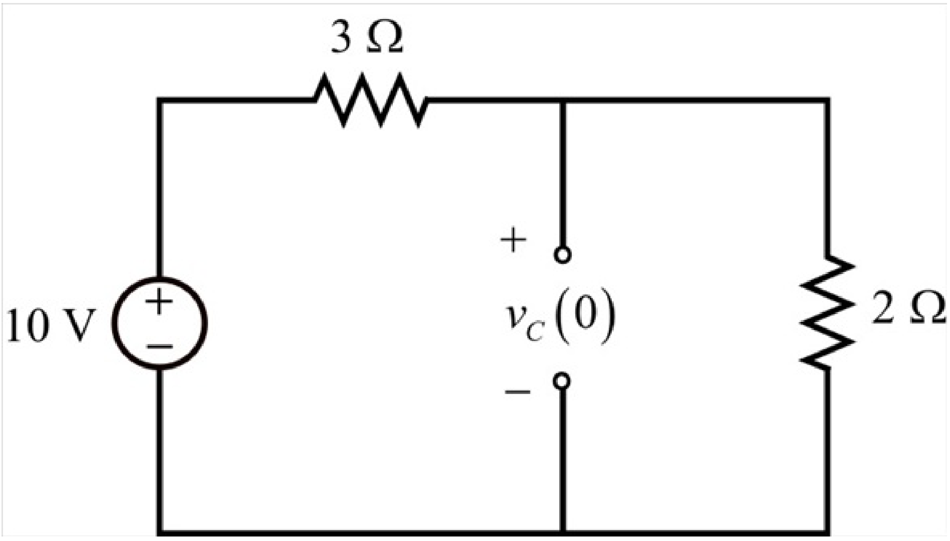


Step-by-step solution

Step 1 of 3

Refer to Figure 7.79 in the text book.

At $t = 0$, the switch is open, but before $t = 0$, the switch is remains in closed position. For DC, capacitor acts like an open circuit. Draw the modified circuit diagram for $t < 0$.



Step 2 of 3

The capacitor voltage is nothing but the voltage across the $2\ \Omega$ resistor. Apply voltage division across $2\ \Omega$ resistor to find the voltage $v_C(0)$.

$$\begin{aligned} v_C(0) &= \left(\frac{2}{2+3}\right)(10\text{ V}) \\ &= \left(\frac{2}{5}\right)(10\text{ V}) \\ &= 4\text{ V} \end{aligned}$$

Therefore, the capacitor voltage just before $t=0$ is 4 V.

Step 3 of 3

Thus, the correct option is $\boxed{(\mathbf{d})}$.